

Customer Behaviour Analysis Report..

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Objective

The objective of this project is to analyze customer transaction data using RFM (Recency, Frequency, Monetary) analysis. The analysis helps identify different customer segments, understand purchasing behavior, improve customer retention, and provide business recommendations.

Dataset Information

- Dataset Name: Customer Behaviour Analysis
- Source: Kaggle

Data Cleaning

➤ The following data cleaning steps were performed:

- Checked for missing values.
- Verified duplicate records.
- Corrected the Purchase Date format.
- Created a clean dataset for analysis.
- Prepared the data for Pivot Tables and RFM analysis.

RFM Analysis

➤ The following metrics were calculated:

- **Recency**

Number of days since the customer's last purchase.

- **Frequency**

Number of purchases made by each customer.

- **Monetary**

Total amount spent by each customer.

➤ Customers were segmented into:

.Champions

.Big Spenders

.Loyal Customers

.Recent Customers

.Others

	A	B	C	D	E	F	G	H	I	J	K
1	Coustomer ID	Frequency	Monetary	Last Purchase Date	Recency	15-09-2023	R Score	F Score	M Score	RFM Score	Customer Segment
2	1	1	3491	19-07-2023	57.71861111		5	1	1	511	Recent Customers
3	2	3	7988	21-11-2022	298.2333681		2	1	2	212	Others
4	3	8	22587	18-06-2023	88.99474537		4	5	5	455	Loyal Customers
5	4	4	8715	12-05-2023	126.0650926		4	2	2	422	Others
6	5	8	12524	28-03-2023	170.9536227		3	5	3	353	Loyal Customers
7	6	6	15517	09-06-2022	462.883125		1	4	4	144	Others
8	7	3	7835	27-02-2022	565.1885532		1	1	2	112	Others
9	8	6	15265	09-05-2023	129.3438773		4	4	4	444	Others
10	9	5	14720	12-05-2023	126.0071528		4	3	3	433	Others
11	10	8	20857	29-01-2023	229.4435069		3	5	5	355	Loyal Customers
12	11	10	23691	01-05-2023	136.7584491		4	5	5	455	Loyal Customers
13	12	6	15378	10-12-2022	279.037037		2	4	4	244	Others
14	13	4	7244	18-06-2023	88.84883102		4	2	1	421	Others
15	14	6	16161	24-03-2022	539.8278819		1	4	4	144	Others
16	15	5	19748	05-06-2023	102.4394213		4	3	5	435	Big Spenders
17	16	5	15246	21-04-2023	147.4159028		3	3	4	334	Others
18	17	9	24042	01-09-2023	13.58564815		5	5	5	555	Champions
19	18	8	23726	22-05-2023	115.5455208		4	5	5	455	Loyal Customers

Dashboard

➤ The dashboard was created to visualize customer behavior.

It contains:

- Customers by Segment
- Revenue by Segment
- Average Recency by Segment



Key Findings

- Most customers belong to the **Others** segment.
- **Champions** contribute the highest revenue.
- **Recent Customers** have the lowest recency and are likely to purchase again.
- **Big Spenders** generate high sales and should be retained.
- RFM analysis successfully identified valuable customer groups.

Actionable Recommendations..

- Reward **Champions** with exclusive discounts and early access to new products.
- Launch a loyalty program for **Loyal Customers**.
- Send personalized offers to **Recent Customers** to encourage repeat purchases.
- Target **Big Spenders** with premium product recommendations and bundle offers.
- Re-engage customers in the **Others** segment through email campaigns and special promotions

The Takeaway

- The Customer Behaviour Analysis project demonstrated how RFM analysis can be used to understand customer purchasing behavior and identify valuable customer segments. The dashboard and recommendations provide practical insights that can support better business decisions and improve customer retention.